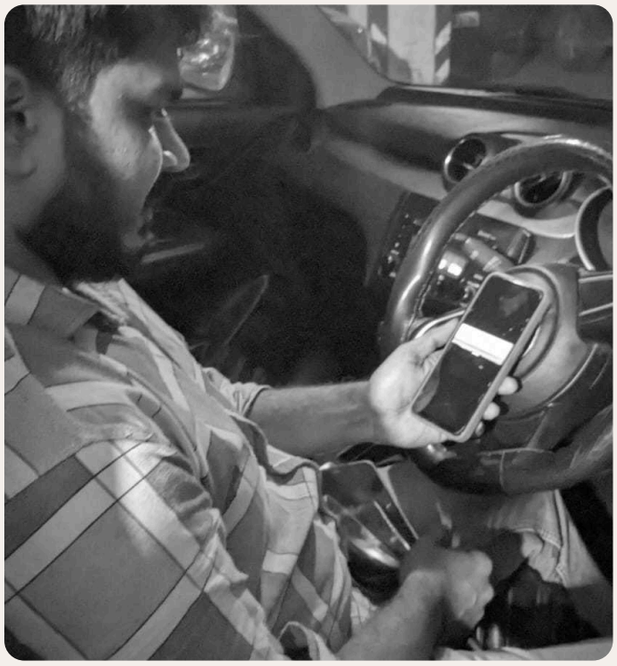
B2B



Highway Hypnosis - Raahi

Team

Shreyash Agare (me),
Aman Sinha,
Sameh Waghmare

Tools

Figma, Fig jam, Vs code, Arduino, Python, Illustrator, Photoshop, DaVinci resolve.

Methods

Design Thinking, Stakeholders mapping, Barrier analysis, Interviews, Determinants Table, Persona creation, Valuation Matrix, Fogg Model, Hook's model, Octalysis, Wireframing, Prototyping, User testing etc

Timeline

2 Months

My Role

I contributed across multiple stages—from **research** to **prototyping**. This included conducting **primary and secondary research**, **synthesizing insights**, **mapping system interactions**, **building the MVP** and **functional prototype**

Research

When the three of us sat down to decide on a project, one thing was clear—our shared love for travel. That naturally led us to explore issues drivers face on Indian highways. One problem stood out: highway hypnosis.

Problem Area

Highway hypnosis or driving hypnosis is when the driver zones out while driving a vehicle without remembering what occurred in that specific period.

"To guide our exploration, we defined clear research objectives—focusing on driver fatigue and rest behavior. After reviewing literature, we conducted 25+ interviews with cab drivers on Sahar Road, often late at night. It was hands-on and revealing. We'd often approach them with, 'Bhaiyya, ek minute baat karni hai, please!'—and their stories shaped our core insights."

Key Insights

Drivers face cumulative fatigue due to poor work conditions and lack of rest.

Cognitive functions decline significantly due to sleep deprivation.

Highway monotony makes the brain drift, leading to dangerous zoning out.

Despite knowing they need breaks, many drivers avoid them due to safety fears, time pressure, or not knowing where to stop.

Finding the Gap

We reframed our challenge into a ‘How Might We’ statement to guide ideation

How Might We

How Might We design a seamless solution for Indian drivers that encourages regular breaks and helps combat highway monotony, ensuring a safer and more engaging experience when driving.

Seamless | Safe | Engaging | Convenient

Ideation & Behavior Design

To move from insight to action, we analyzed driver behavior

- Recognized Current Behaviors
- Mapped Desired Behaviors
- Conducted Barrier Analysis: Why don’t drivers stop?
- Used Determinants Table to pinpoint behavioral bottlenecks.
- Evaluated concepts using a solution matrix.

Raahi

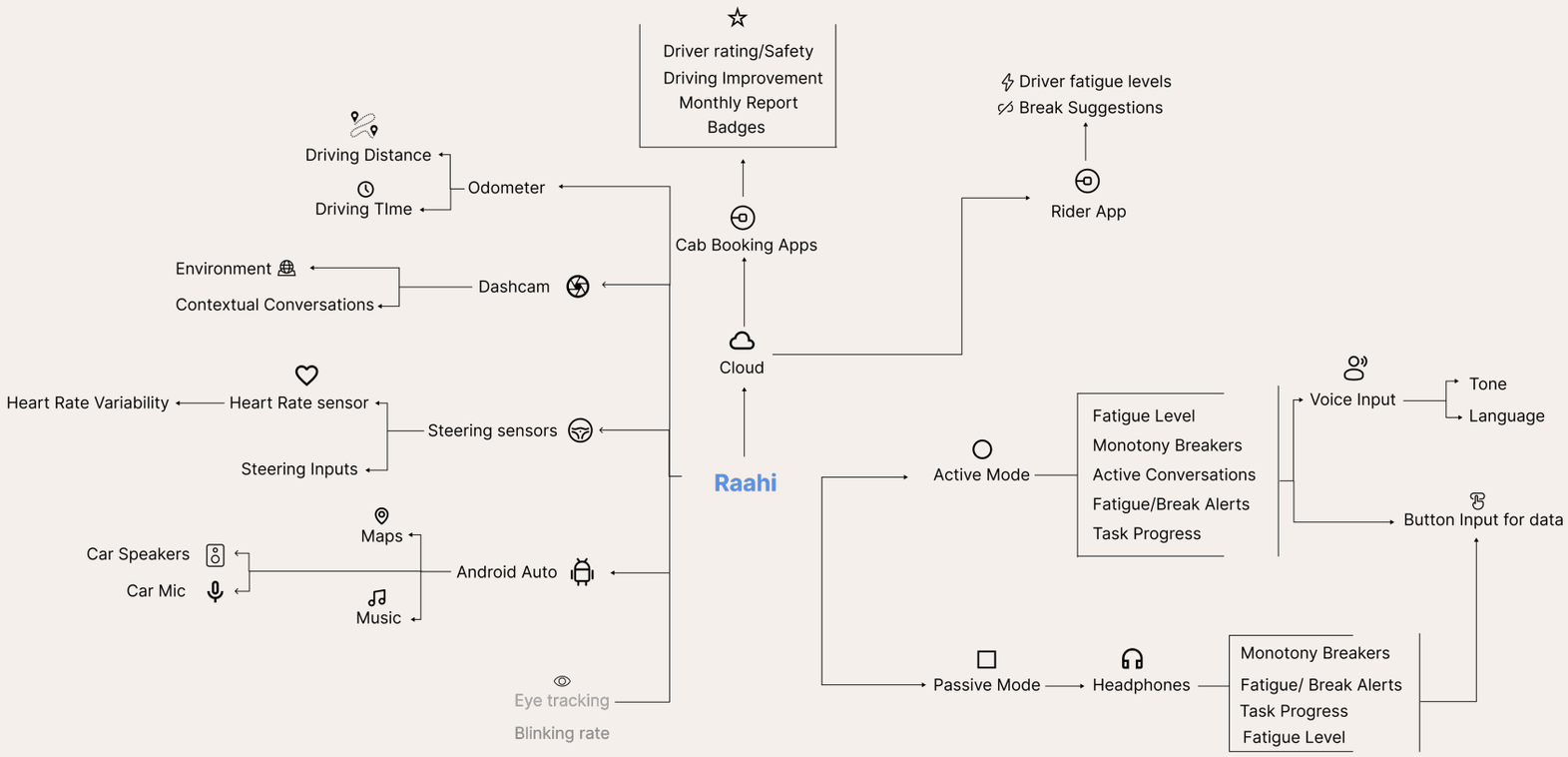
Why Voice?

- Keeps eyes on the road—no screen distractions like HUD.
- Encourages breaks gently, overcoming common driver excuses.
- Fits most vehicles, easy to install and low-cost.
- Becomes a conversation bridge between driver and passenger.

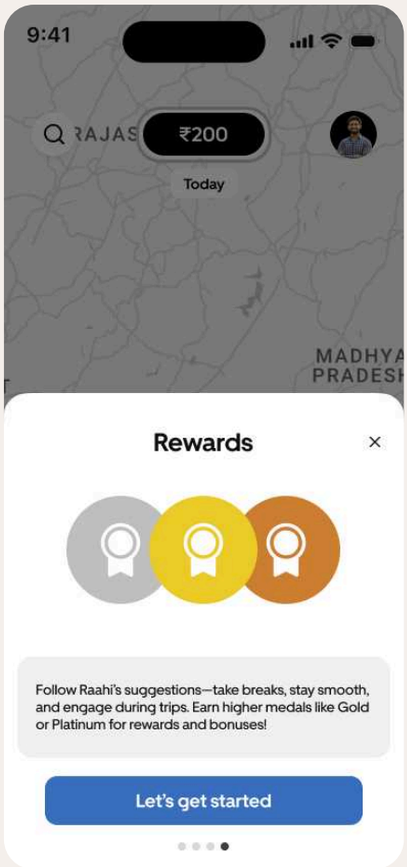
System Features

- Voice Prompts: Personalized break reminders.
- Gamified Rewards: Medals instead of grades to motivate.
- Adaptive Alerts: Visual (HUD) → Tactile (vibration) → Audio (urgent voice).
- Mode-Specific:
- Highway Mode: Shows fatigue data, rest stops, gamified tasks.
- City Mode: Minimal alerts, subtle fatigue cues only.

System Architecture



Our Solution



Product costing

Independent

Microphone Array	Rs. 200-500
4 Speakers	Rs. 500-700
Processor	Rs. 350-450
RAM (usually in processor)	Rs. 300
DSP Chips	Rs. 300-500
LED Strips	Rs. 100-250
Power Supply	Rs. 150-300

Cost:
2.5k - 3.5k

Business model

Value Proposition

- Keeps drivers safe, refreshed, and alert
- Boosts earnings by 10–30% via habits & rewards
- Unique edge: AI fatigue detection + gamification

Revenue Streams

- One-time hardware sales
- Licensing to OEMs
- 2% of Raahi-enabled trip earnings
- Commission from rest stop partnerships
- Data monetization (insurance, govts.)

Key Partnerships

- Uber (distribution)
- Auto manufacturers (pre-installation)
- Fuel stations/rest stops (break stops)
- Governments & insurers (data use)

User Testing & Behavior Modeling

Current Behavior

1. Drivers unaware of fatigue until too late.
2. Drivers neglect breaks due to passengers
3. Long drives lead to disengagement.
4. Current systems respond after fatigue appears.

Desired Behavior

1. Drivers recognize fatigue early with real-time sensing.
2. System prompts breaks and engages passengers into the decision
3. Gamification and energy indicators keep drivers alert.
4. System tries prevention of Fatigue.

We tested 'Rahi' with real drivers and refined it based on feedback. We also validated it using behavioral frameworks.

Frameworks Used (Behavioral):

Hook Model | Habit Loop | Fogg's Behavior Model | Octalysis Framework

UT Rating ★★★★☆

What Worked

- Break reminders improved safety
- Medals felt motivating
- Quest bonuses were engaging
- Voice assistant helped initiate breaks

What Didn't Work

- Too many notifications
- Letter grades felt judgmental
- Needed tailored profiles for different drivers
- Clarity on how fatigue is calculated was lacking

Conclusion

We believe 'Rahi' isn't just a product—it's a companion. A guide that keeps drivers safe, alert, and engaged. Future work includes improving fatigue detection using multi-sensor input and refining alert personalization

Want better understanding watch these videos